Deep Learning Fundamentals · Chapter 1: The Artificial Neuron

An Artificial Neuron (or Perceptron) is the fundamental atomic building block of deep neural networks. E

The weighted sum is subsequently passed through a non-linear activation function, f(z), allowing the ne

Deep Learning Fundamentals · Chapter 2: Backpropagation and Gradient Descent

Backpropagation is an algorithm for calculating the gradient of the loss function with respect to the weig

1. Forward Pass: Input features propagate forward through network layers to produce the prediction ou
2. Loss Computation: A scalar loss, such as Mean Squared Error (MSE) or Cross-Entropy Loss, quanti
3. Backward Pass: Gradients delta_L / delta_w are propagated backwards layer-by-layer.
4. Weight Update: Gradient Descent updates the parameters according to w_new = w_old - alpha * gra

Deep Learning Fundamentals · Chapter 3: Activation Functions and Vanishing Gradients

The Vanishing Gradient problem occurs when gradients shrink exponentially as backpropagation propa

- Sigmoid Activation: Maps inputs to (0, 1). Its derivative saturates to near-zero for large absolute value
- Tanh Activation: Zero-centered with output range (-1, 1), but still suffers from saturation and vanishing
- ReLU (Rectified Linear Unit): Defined as f(z) = max(0, z). Its derivative is constant (1) for all positive in